CIRCUIT A

1) Resistance total

$$R_T = R_1 + R_2 + R_3$$

$$= 20 + 30 + 50$$

$$= 100\,\Omega$$

2) Total current

$$I = \frac{V}{R_t}$$

$$= \frac{125}{100}$$

$$= 1{\cdot}25\,A$$

3) Current through each resistor

Circuit in series hence current is the same for each resistor.

$1{\cdot}25\,A$

4) Voltage drop across each resistor

$$V = IR$$

$20\,\Omega$ resistor $= 1{\cdot}25 \times 20 = 25\,V$

$30\,\Omega$ resistor $= 1{\cdot}25 \times 30 = 37{\cdot}5\,V$

$50\,\Omega$ resistor $= 1{\cdot}25 \times 50 = 62{\cdot}5\,V$

5. Power dissipated by each resistor

$P = VI$

$20\,\Omega$ resistor $= 25 \times 1.25 = 31.25W$

$30\,\Omega$ resistor $= 37.5 \times 1.25 = 46.88W$

$50\,\Omega$ resistor $= 62.5 \times 1.25 = 78.13W$

CIRCUIT B

1) Total resistance

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$= \frac{1}{0.05 + 0.01 + 0.02}$$

$$= \frac{1}{0.08}$$

$$= 12.5\,\Omega$$

2) Total current

$$I = \frac{V}{R_t} = \frac{125V}{12.5\,\Omega}$$

$$= 10A$$

3) Current through each resistor

$20\,\Omega$ resistor $= \frac{125V}{20\,\Omega} = 6.25A$

$100\,\Omega$ resistor $= \frac{125V}{100\,\Omega} = 1.25A$

$50\,\Omega$ resistor $= \frac{125V}{50\,\Omega} = 2.5A$

4) Voltage drop across each resistor.

Circuit in parallel so each resistor receives full voltage.

$$V_{20\Omega} = V_{100\Omega} = V_{50\Omega} = 125\,V$$

5) Power dissipated through each resistor

$$P_{20\Omega} = 125V \times 6.25A = 781.25W$$

$$P_{100\Omega} = 125V \times 1.25A = 156.25W$$

$$P_{50\Omega} = 125V \times 2.5A = 312.5W$$